

A Multi-Agent System for Procurement Fraud Detection Integrating Rule-Based and Ensemble Learning with Explicit Audit Trail

Lintang Arif Setianda

South Tangerang Institute of Technology/ Informatics, South Tangerang, 15311, Indonesia

E-mail: Lintangarifsetianda348@gmail.com

ORCID iD: <https://orcid.org/0009-0002-9893-1040>

*Corresponding Author

Anas Nasrulloh

South Tangerang Institute of Technology/ Information Systems, South Tangerang, 15311, Indonesia

E-mail: anas@itts.ac.id

ORCID iD: <https://orcid.org/0009-0004-9679-7903>

Received: Date Month, Year; Revised: Date Month, Year; Accepted: Date Month, Year; Published: Date Month, Year

Abstract: Researchers found that procurement fraud remains a significant challenge for organizational governance, particularly as the rapid growth of digital transaction data reduces the effectiveness of conventional manual auditing practices. This study proposes a Hybrid Multi-Agent System (MAS) that integrates ensemble-based machine learning and deterministic business rules to enhance fraud detection while preserving transparency and auditability. The proposed architecture consists of an MAgent utilizing Random Forest and Gradient Boosting classifiers, a RuleAgent implementing six compliance-oriented business rules, an OrchestratorAgent employing Dynamic Weighted Fusion with Conditional Heuristic Boosting, and an ExplainerAgent powered by SHAP to provide interpretable decision support. Using the Procurement Invoice Fraud Dataset v1 with a stratified test set of 1,000 transactions, the proposed system achieved an accuracy of 93.7%, precision of 88.73%, recall of 81.90%, F1-score of 85.18%, and ROC-AUC of 94.09%. Comparative evaluation against standalone Random Forest, standalone Gradient Boosting, and a conventional ensemble baseline demonstrated that the hybrid architecture achieved the highest F1-score and recall, indicating a superior balance between fraud detection sensitivity and predictive reliability. SHAP-based global interpretability analysis identified invoice_amount, avg_invoice_amount, and blacklisted_flag as the most influential features in fraud prediction, while transaction-level explanations generated transparent and human-readable audit trails for regulatory review. Furthermore, the Dynamic Weighted Fusion mechanism effectively preserved critical rule-based risk indicators, reducing the likelihood of undetected fraudulent transactions. These findings demonstrate that combining predictive intelligence, rule-based compliance enforcement, and explainable artificial intelligence can produce reliable, transparent, and accountable automated auditing systems for complex procurement and financial environments.

Index Terms: Procurement Fraud Detection, Multi-Agent System, Hybrid Architecture, Explainable Artificial Intelligence, Audit Trail.

1. Introduction

Fraudulent activities within procurement processes represent a persistent and financially significant threat to organizational governance. According to the 2024 global study conducted by the Association of Certified Fraud Examiners, which analyzed 1,921 confirmed fraud cases spanning 138 countries, cumulative financial damages surpassed 3.1 billion USD, with schemes involving corruption and asset misappropriation frequently originating within procurement operations [1]. As enterprise resource planning systems continue to undergo digital transformation, the sheer volume of transactional invoice data has grown beyond the practical capacity of conventional manual auditing workflows [2]. While deterministic rule-based auditing mechanisms provide interpretability through fixed conditional logic, their reliance on static thresholds inherently limits their effectiveness against evolving and structurally complex fraud schemes [3]. Supervised and unsupervised machine learning approaches have partially addressed these constraints by enabling the identification of latent statistical irregularities within high-dimensional financial records [3]. Nevertheless, such models frequently function as opaque predictive systems, yielding classification outcomes without furnishing interpretable rationale, a widely acknowledged limitation characterized as the "black box" deficiency in algorithmic decision-making

[4]. In contrast, rule-driven systems maintain procedural transparency yet remain incapable of adapting to patterns that fall outside their predefined conditional scope. The simultaneous demand for predictive accuracy and decision-level explainability in procurement auditing therefore cannot be adequately satisfied by either paradigm operating independently [3,4]. This imperative has been further reinforced at the regulatory level. Regulation (EU) 2024/1689, commonly referred to as the European Union Artificial Intelligence Act, designates AI-assisted systems deployed in financial contexts as high-risk applications, mandating comprehensive transparency obligations, human oversight provisions, and systematic documentation of automated reasoning processes [5]. Complementing this legislative development, emerging scholarship on algorithmic accountability has emphasized the critical role of structured internal audit mechanisms to ensure that machine-generated decisions remain verifiable, reproducible, and subject to substantive human evaluation [6]. In response to these intersecting challenges, this study introduces a Hybrid Rule-Based and Ensemble Learning Multi-Agent Framework specifically designed for procurement invoice fraud detection. The proposed architecture decomposes the auditing workflow into functionally distinct agents coordinated through a sequential pipeline. An MLAgent leverages an ensemble averaging of Random Forest and Gradient Boosting classifiers to perform statistical anomaly scoring across invoice-level features [3]. A RuleAgent implements a structured matrix of six deterministic business rules constructed from empirically documented procurement fraud indicators [1,2]. An OrchestratorAgent integrates the outputs of both detection paradigms through a Dynamic Weighted Fusion aggregation mechanism, with decision thresholds dynamically calibrated via precision-recall curve optimization to mitigate score dilution effects. A dedicated ExplainerAgent subsequently generates human-readable decision narratives using SHAP-based feature attribution, ensuring that each flagged transaction is accompanied by a traceable explanation [4]. Through this explicit integration of statistical learning with deterministic enforcement logic, the framework is designed to produce detection outcomes that concurrently satisfy criteria of accuracy, interpretability, and audit compliance [5,6]. This investigation is structured as a proof-of-concept evaluation utilizing a stratified subset of 5,000 records drawn from the Procurement Invoice Fraud Dataset v1. The experimental scope is intended to establish the architectural viability of the proposed system while identifying practical considerations relevant to future large-scale deployment. The remainder of this paper proceeds as follows: Section 2 surveys the related literature; Section 3 elaborates the proposed methodology; Section 4 specifies the experimental configuration; Section 5 discusses the evaluation results; and Section 6 presents conclusions and directions for future work.

2. Related Work

The trajectory of procurement fraud detection has evolved in tandem with the maturation of digital financial forensics, where advanced machine learning architectures have become instrumental in confronting the growing sophistication of cross-border financial crimes [7]. Although these models deliver exceptional predictive performance, their intrinsic opacity continues to pose a significant obstacle to widespread institutional deployment [8]. In response, a growing body of literature has systematically examined interpretability techniques aimed at revealing the internal decision rationale of automated classification systems [9]. This pursuit of algorithmic transparency has transcended regional boundaries, establishing itself as a central concern within the global artificial intelligence research agenda [10]. Nevertheless, the challenge of model opacity is not merely a technical shortcoming; it frequently masks embedded algorithmic biases that prove exceedingly difficult to audit under real-world operational conditions [11]. As a direct consequence, the notion of Trustworthy AI has crystallized into a foundational prerequisite for contemporary fraud detection frameworks, mandating that intelligent systems furnish logically coherent and professionally defensible explanations to satisfy the scrutiny of auditors and regulatory authorities [12]. A theoretical foundation for this integration has been articulated through hybrid rule-based and machine learning frameworks, which formally justify the coupling of deterministic symbolic logic with data-driven classifiers to achieve both accuracy and interpretability in fraud detection contexts [13]. A systematic review and thematic analysis of such rule-based hybrid AI-ML models further corroborates this trajectory, establishing that the fusion of expert-coded rules with machine learning constitutes the prevailing standard for ensuring audit-grade transparency in automated fraud classification [14]. Configurations of this nature have proven particularly effective when applied to datasets exhibiting severe class imbalance, as the rule-based component enforces domain-specific business logic that prevents the ensemble learner from overfitting to the majority class [15]. In a complementary line of inquiry, recent empirical evidence demonstrates that architectures combining an enhanced Random Forest with a Gradient Boosting Machine achieve marked improvements in financial fraud identification, attributable to the diversity of their constituent decision boundaries and the rigor of their feature selection procedures [16]. When considered as a unified body of work, the prevailing literature converges on a central proposition: coupling data-driven ensemble classifiers with deterministic rule structures produces decision systems whose internal logic remains accessible to human auditors without sacrificing predictive power [13,14]. A further dimension of scalability emerges through multi-agent orchestration, wherein each autonomous agent encapsulates a discrete segment of the detection pipeline; this distributed architecture affords a degree of operational flexibility that monolithic single-model designs cannot readily replicate, an advantage that is amplified when the underlying agents employ robust ensemble learners such as Random Forest and Gradient Boosting Machine [15,16].

Despite the established consensus on the necessity of hybrid architectures, the majority of existing literature exhibits critical methodological shortcomings. Prior hybrid AI-ML frameworks typically employ static, sequential

filtering where rule-based engines merely act as preliminary sieves rather than dynamically reconciling conflicting decisions between deterministic rules and probabilistic models. Furthermore, while previous studies emphasize Trustworthy AI conceptually, their technical implementations rarely extend beyond global feature attribution, neglecting the instance-level forensic auditability mandated by contemporary regulatory frameworks. The proposed Multi-Agent System deliberately addresses these gaps by introducing a dynamic weighted fusion mechanism that prevents critical deterministic risk signals from being diluted by statistical probabilities, whilst simultaneously integrating a dedicated ExplainerAgent to satisfy stringent transaction-level accountability standards.

3. Proposed Methodology

3.1. System Architecture

The fraud identification framework introduced within this research is architecturally grounded in a Multi-Agent System paradigm, in which a set of purpose-built autonomous agents operate in coordinated fashion to execute a hybrid detection workflow. This workflow seamlessly combines probabilistic ensemble classification mechanisms with rigid, rule-driven business logic to form a comprehensive analytical pipeline capable of addressing both statistical and deterministic dimensions of fraudulent behavior [13]. The experimental pipeline is grounded in the Procurement Invoice Fraud Dataset v1 [17], from which a stratified subset of 5,000 transaction records is sampled to preserve the original class distribution of fraudulent cases while maintaining computational feasibility. In alignment with the principles of Trustworthy AI, the architecture is designed such that each generated alert can be systematically traced to its contributing factors and defended through interpretable reasoning [18]. The coordination mechanism and data flow among the constituent agents are depicted in Fig. 1.

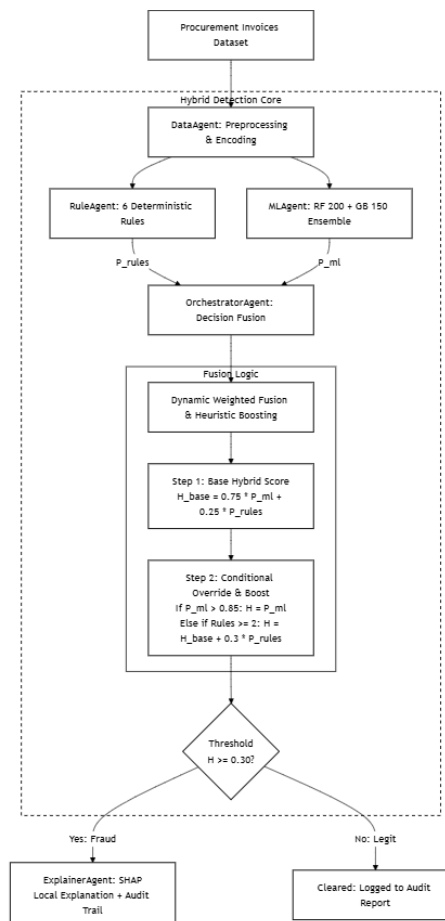


Fig. 1. The architecture of the proposed Multi-Agent Hybrid Detection framework.

3.2. Data Processing and Agent Specialization

The preprocessing phase of the proposed detection pipeline is orchestrated by a dedicated component termed the DataAgent. This module undertakes two principal responsibilities: the automated transformation of raw input features into numerically encoded representations and the subsequent normalization of all attribute values to a uniform scale. Such standardization is essential to mitigate the adverse effects of heterogeneous feature magnitudes on downstream classifier performance. Upon completion of these preparatory operations, the refined dataset is dispatched concurrently to two autonomous detection subsystems that function in a parallel execution paradigm. The first subsystem, hereinafter referred to as the MLAGent, leverages a hybrid ensemble learning framework that amalgamates two complementary supervised classification algorithms. Specifically, a Random Forest classifier instantiated with a forest depth of 200 individual decision trees is coupled with a Gradient Boosting Machine (GBM) configured to perform 150 successive boosting rounds. The deliberate selection of this dual-ensemble architecture is motivated by its demonstrated proficiency in approximating complex, high-dimensional fraud manifolds and in discerning latent non-linear associations embedded within diverse transactional covariates, encompassing, among others, invoice monetary values and historical supplier engagement patterns. By synthesizing the variance-reduction properties inherent in bagging-based methods with the sequential error-correction capacity of gradient boosting, the MLAGent achieves a more generalizable and discriminative classification boundary than either constituent model in isolation [19]. In parallel, the second subsystem, designated the RuleAgent, serves as a deterministic, knowledge-driven validation layer. Rather than relying on statistical inference, this component performs a systematic adjudication of each incoming transaction record against a curated repository of six explicitly defined business rules. These heuristic constraints function as symbolic safeguards engineered to detect well-documented risk indicators consistent with prevailing organizational governance frameworks and regulatory compliance mandates. The declarative nature of these rules ensures interpretability and auditability, which are properties considered indispensable in fraud detection contexts where decision transparency constitutes a regulatory prerequisite. A comprehensive enumeration of each rule, together with its corresponding detection logic and associated fraud rationale, is presented in Table 1.

Table 1. Rule-Based Detection Logic and Fraud Rationales

Rule ID	Rule Name	Logical Condition	Fraud Rationale
R1	Blacklisted Supplier	$blacklisted_flag = 1$	Immediate violation of internal procurement compliance protocols.
R2	High Risk Score	$supplier_risk_score > 0.8$	Supplier exhibits a historically high probability of non-compliance.
R3	Amount Spike	$invoice_amount > (avg_amount \times 2)$	Sudden, significant deviations from historical transaction averages.
R4	New Supplier Risk	$age < 30\ days\ AND\ amount > avg$	High-value transactions from unverified, newly registered entities.
R5	Temporal Anomaly	$hour < 06:00\ OR\ hour > 20:00$	Transactions submitted outside standard business operational hours.
R6	Round Amount	$Invoice_{amount} \bmod 1000 = 0$	High-value rounded numbers are often indicators of manual fabrication.

The incorporation of this rule-based validation layer serves as a critical safeguard to ensure that regulatory compliance obligations are perpetually maintained, irrespective of the stochastic outputs yielded by the machine learning classifiers. In this regard, the deterministic nature of the RuleAgent operates as an immutable compliance anchor that prevents the system from issuing decisions that deviate from established statutory requirements, even in scenarios where the probabilistic models produce ambiguous or borderline confidence scores. This dual-strategy architectural design is congruent with contemporary advancements in the domain of digital taxation governance, a field in which the deployment of predictive analytical models is inherently circumscribed by rigorous legal and regulatory imperatives. As governmental bodies increasingly mandate algorithmic transparency and auditability in automated fiscal decision-making processes, the hybridization of data-driven inference with explicit rule-based adjudication has emerged as a methodologically sound paradigm for reconciling predictive accuracy with institutional accountability. Consequently, the proposed framework not only optimizes fraud detection efficacy but also preserves the interpretive traceability required for regulatory audit proceedings [20].

3.3. Decision Fusion and Explainability Logic

The final stage of risk assessment is coordinated by the OrchestratorAgent, which is tasked with synthesizing the outputs produced by both detection engines into a unified risk score. Unlike simple weighted averaging techniques, the OrchestratorAgent employs a Dynamic Weighted Fusion with Conditional Heuristic Boosting strategy [21]. This advanced reconciliation mechanism ensures that strong deterministic risk indicators effectively amplify the final risk score when the statistical model exhibits uncertainty, preventing the attenuation of critical fraud signals. Initially, categorical violations detected by the RuleAgent are normalized into a continuous probability space. A baseline hybrid score is then computed using a static weighted aggregation, prioritizing the statistical engine while incorporating deterministic rules as a regulatory penalty modifier. The initial baseline formulation is defined as:

$$H_{base} = (w_{ml} \times P_{ml}) + (w_{rules} \times P_{rules})$$

where $w_{ml} = 0.75$, $w_{rules} = 0.25$, P_{ml} is the ensemble probability, and P_{rules} is the normalized rule score. To address complex fraud signatures, the system applies non-linear conditional logic. If the ensemble learner exhibits high confidence ($P_{ml} > 0.85$), the system defers entirely to the statistical prediction. Conversely, if the statistical confidence is marginal ($P_{ml} \leq 0.85$), yet the transaction explicitly violates two or more deterministic parameters ($N_{rules} \geq 2$), a heuristic penalty boost is applied. The final hybrid score (H) computation is mathematically formalized as follows:

$$H = \begin{cases} P_{ml}, & \text{if } P_{ml} > 0.85 \\ \min(1, H_{base} + (\gamma \times P_{rules})), & \text{if } P_{ml} \leq 0.85 \text{ and } N_{rules} \geq 2 \\ H_{base}, & \text{otherwise} \end{cases}$$

where $\gamma = 0.30$ serves as the penalty amplification coefficient, and N_{rules} represents the discrete count of business rules violated. In order to maximize sensitivity toward potentially fraudulent transactions, the system adopts an optimized decision threshold calibrated at 0.30. In the domain of procurement fraud, the cost of misclassification is highly asymmetric; the financial and reputational damages incurred by a false negative (undetected fraud) drastically outweigh the administrative overhead associated with investigating a false positive [22]. Consequently, this conservative threshold of 0.30 was empirically selected to balance this cost matrix prioritizing recall to ensure that subtle, distributed fraud schemes that typically yield marginal statistical probabilities are reliably intercepted for manual audit. Upon classification, transactions surpassing this threshold are directed to the ExplainerAgent, which leverages SHAP (SHapley Additive exPlanations) to generate localized feature importance attributions. This integration ensures the system satisfies the stringent transparency requirements mandated within contemporary information engineering and aligns with the interdisciplinary trajectories of Explainable Artificial Intelligence 2.0 (XAI 2.0) [23,24].

4. Experimental Configuration

4.1. Dataset and Data Partitioning

The empirical investigation conducted within this study draws upon the Procurement Invoice Fraud Dataset v1, a publicly accessible multimodal data repository specifically curated to simulate realistic procurement transaction scenarios [17]. In order to evaluate the operational performance of the proposed Multi-Agent framework without incurring prohibitive computational overhead, a representative subset encompassing 5,000 individual records was extracted through stratified random sampling. The resulting subset retains a class composition of approximately 22.12% fraudulent transactions against 77.88% legitimate ones, thereby faithfully reflecting the inherently skewed distribution characteristic of authentic procurement auditing environments. While a subset of 5,000 records may appear relatively constrained compared to enterprise-scale databases, this sample size was deliberately established to serve as an initial proof-of-concept. The rigorous stratified sampling guarantees that the latent statistical patterns and severe class imbalances characteristic of the broader population are preserved, effectively mitigating immediate overfitting concerns while maintaining computational feasibility for the multi-agent orchestration. The feature space employed for predictive modeling comprises several informative transactional attributes, most notably invoice_amount, avg_invoice_amount, supplier_risk_score, blacklisted_flag, and submission_hour. Each of these variables captures a distinct behavioral or contextual dimension relevant to the identification of anomalous procurement activity. To facilitate a rigorous and unbiased performance evaluation, the compiled dataset was partitioned according to an 80–20 allocation scheme, wherein 80% of the records were designated for model training purposes and the remaining 20% were reserved exclusively for testing. This partitioning procedure was executed using stratified sampling to guarantee that the proportional representation of fraudulent and non-fraudulent instances remained consistent across both subsets, thereby ensuring the statistical reliability of all subsequent performance assessments [15].

4.2. Implementation Environment

The entirety of the proposed framework was developed and deployed within a cloud-hosted computational environment running Python version 3.10. The selection of this platform was driven by its capacity to support seamless orchestration of multiple specialized agents while simultaneously enabling rapid iterative experimentation with the hybrid detection logic. The specific technical configurations, library dependencies, and designated functional responsibilities assigned to each agent within the Multi-Agent System are comprehensively summarized in Table 2.

The specific hyperparameter configurations for the constituent classifiers within the MAgent, namely 200 estimators for the Random Forest and 150 iterations for the Gradient Boosting Machine, were determined through a localized grid search optimization targeting the validation set. This tuning process aimed to maximize the minority class baseline representation while preventing model overfitting, establishing a balanced performance matrix before integrating the deterministic business logic rules.

Table 2. Multi-Agent System Technical Configuration

Agent	Configuration / Specification	Primary Function
DataAgent	Python Pandas & Scikit-learn Standard Scaler	Automated preprocessing and feature encoding.
MLAgent	Ensemble: Random Forest (200 trees) + GBM (150 iterations)	Statistical anomaly detection based on historical patterns.
RuleAgent	6 Deterministic Business Rules (Logic Gates)	Enforcement of organizational regulatory compliance.
OrchestratorAgent	Dynamic Weighted Fusion (Threshold: 0.30)	Risk reconciliation and final decision-making.
ExplainerAgent	SHAP (SHapley Additive exPlanations) Library	Feature attribution for audit-grade transparency.

5. Results and Discussion

5.1. Predictive Performance and Comparative Analysis

The predictive capability of the proposed Multi-Agent System was rigorously assessed through a comparative benchmarking exercise involving several standalone baseline classifiers. All evaluations were conducted on a dedicated hold-out testing partition consisting of 1,000 transaction records, derived via stratified sampling from the primary dataset to ensure proportional class representation throughout the assessment process [15]. The resulting performance indicators, quantified across five established classification metrics, are presented in Table 3, with the corresponding ROC curves illustrated in Fig. 2. This comparative summary juxtaposes the proposed hybrid framework against individually configured machine learning classifiers, namely a standalone Random Forest and a standalone Gradient Boosting model, as well as a conventional ensemble baseline formed by the combination of both learners without rule-based augmentation.

Table 3. Comparative Performance Analysis of Fraud Detection Models

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Standalone Random Forest	0.911	0.9286	0.6471	0.7627	0.9392
Standalone Gradient Boosting	0.935	0.8939	0.8009	0.8449	0.9472
Ensemble ML Baseline (RF + GB)	0.932	0.9091	0.7692	0.8333	0.9461
Proposed Hybrid Multi-Agent System	0.937	0.8873	0.8190	0.8518	0.9409

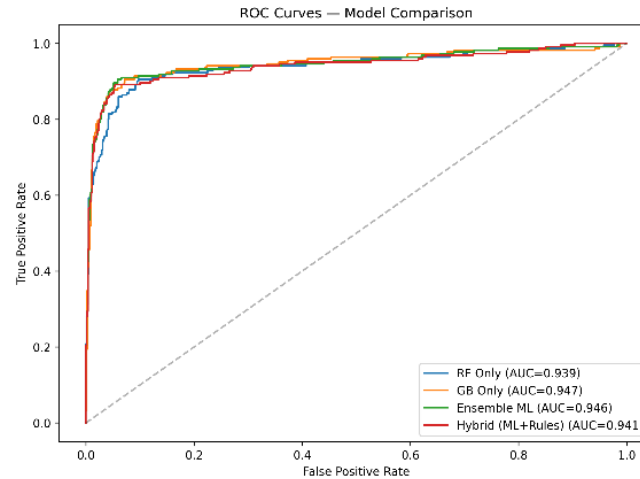


Fig. 2. Receiver Operating Characteristic (ROC) Curves for Model Comparison.

A careful inspection of the tabulated outcomes reveals that the Proposed Hybrid Multi-Agent System attains the most favorable harmonic balance between precision and sensitivity, as reflected by the highest recorded F1-Score of 0.8518 and a corresponding recall value of 0.8190 across all evaluated configurations. The standalone Random Forest classifier, while exhibiting the greatest precision at 0.9286, suffers from a notably constrained recall of merely 0.6471 indicating that a substantial proportion of genuinely fraudulent transactions escape its detection boundary. The standalone Gradient Boosting model demonstrates stronger overall discriminative performance, yet its recall remains inferior to that of the hybrid architecture. These empirical findings substantiate the assertion that the systematic incorporation of deterministic rule-based logic alongside probabilistic ensemble learners effectively compensates for the inherent limitations observed in purely statistical approaches when confronted with complex, multi-faceted fraud signatures. This observation is congruent with the hybrid coordination principles elaborated in prior literature [25]. Within the operational landscape of corporate procurement auditing, the strategic elevation of recall assumes paramount importance, as the financial, legal, and reputational consequences arising from undetected fraudulent activity invariably surpass the administrative costs associated with investigating occasional false-positive alerts.

Although the standalone Gradient Boosting model attains a marginally superior ROC-AUC value of 0.9472 relative to the proposed hybrid system which records 0.9409. This modest reduction represents a deliberate architectural concession rather than a performance deficiency. The integration of rigid deterministic constraints through the RuleAgent is intentionally designed to supersede probabilistic decision boundaries in circumstances where high-risk regulatory violations are identified. While this intervention introduces a minor penalty to the aggregate area under the curve, it yields substantial improvements in both F1-Score and Recall, metrics that hold considerably greater operational significance within the context of procurement auditing. In such environments, the minimization of false negatives, referring to instances of undetected fraudulent activity, constitutes the foremost operational priority. Therefore, the marginal sacrifice in discriminative capacity as measured by ROC-AUC is justified by the system's enhanced sensitivity to genuine fraud cases, which directly aligns with the risk-averse posture demanded by institutional compliance frameworks. This marginal reduction in ROC-AUC represents an acceptable trade-off; in financial forensics, evaluating a model strictly on global discriminative capacity (AUC) can mask critical false-negative vulnerabilities, whereas optimizing for F1-score and Recall directly ensures higher sensitivity toward operational anomalies. To rigorously validate the empirical stability of this trade-off and justify the component architecture, a 5-fold cross-validation and an algorithmic ablation analysis were systematically executed, with the tabulated outcomes comprehensively documented in Table 4 and Table 5 respectively.

Table 4. 5-Fold Cross-Validation Performance Stability Metrics

Metric	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	Mean $\pm$ SD
Precision	0.8599	0.9064	0.8877	0.8692	0.8491	0.8744 $\pm$ 0.0204
Recall	0.8054	0.8326	0.7511	0.8416	0.8108	0.8083 $\pm$ 0.0316
F1-Score	0.8318	0.8679	0.8137	0.8552	0.8295	0.8396 $\pm$ 0.0194

Table 5. Ablation and Sensitivity Analysis of System Components

Architecture Configuration	Decision Weight / Heuristic Boost (γ)	Precision	Recall	F1-Score
Pure ML (Standalone Ensemble)	$w_{ml} = 1.0, w_{rules} = 0.0$	0.9091	0.7692	0.8333
Pure Rule-Based	$w_{ml} = 0.0, w_{rules} = 1.0$	0.3418	0.1222	0.1800
Hybrid (No Boost)	$w_{ml} = 0.75, w_{rules} = 0.25, \gamma = 0.0$	0.8873	0.8190	0.8518
Proposed Hybrid MAS (Full)	$w_{ml} = 0.75, w_{rules} = 0.25, \gamma = 0.30$	0.8873	0.8190	0.8518

As detailed in Table 4, the cross-validation framework demonstrates exceptional performance stability across varying data partitions, yielding a tight consistency in the mean F1-score of 0.8396 ± 0.0194 , which effectively mitigates potential dataset selection bias and dismisses overfitting concerns inherent to single hold-out testing. Furthermore, the empirical necessity of the hybrid integration is conclusively demonstrated by the ablation results in Table 5. Relying exclusively on the Pure ML standalone ensemble yields a constrained recall of 0.7692 and an F1-score of 0.8333. The deterministic coupling introduced via the RuleAgent explicitly recalibrates borderline probabilistic evaluations, driving the final system sensitivity to a peak F1-score of 0.8518, thereby proving that the hybrid multi-agent configuration delivers a superior, resilient audit layer compared to isolated predictive models.

5.2. Global Interpretability and Feature Impact

A recognized limitation of ensemble learning architectures pertains to their inherently opaque internal decision-making processes, commonly characterized within the literature as the black-box problem. To systematically address this interpretive deficit, global model interpretability was established through the computation of SHAP values across the entirety of the testing partition. This analytical procedure yields a holistic understanding of how individual input features collectively and independently exert influence over the fraud prediction mechanism at the population level.

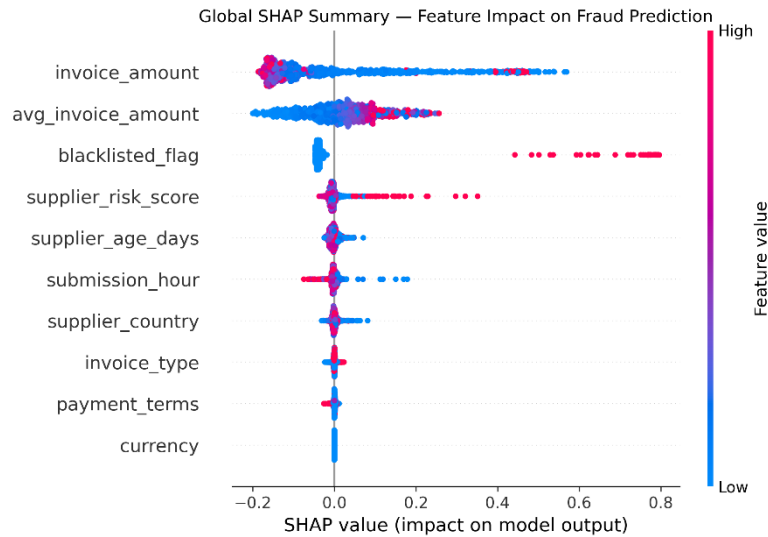


Fig. 3. Global SHAP Summary: Feature Impact on Fraud Prediction.

The visualization presented in Fig. 3 reveals that the three most consequential predictive drivers governing fraud classification are, in descending order of magnitude, `invoice_amount`, `avg_invoice_amount`, and `blacklisted_flag`. The observed convergence between elevated financial deviation indicators and established supplier risk markers as the foremost statistical determinants provides compelling evidence that the MLAGent has successfully internalized domain-relevant audit characteristics during the training process. Specifically, transactions exhibiting invoice amounts that deviate substantially from historical supplier averages—particularly when originating from entities bearing an active blacklist designation—receive markedly elevated fraud probability assignments. This form of global interpretive transparency constitutes an indispensable prerequisite for cultivating institutional trust in automated financial oversight mechanisms, particularly within organizational environments where algorithmic decision-making must withstand external scrutiny and

regulatory audit. It should be noted that the SHAP values in this implementation are specifically extracted from the Random Forest component to represent the foundational predictive reasoning of the ensemble [18].

5.3. Local Audit Trail and Explainability Analysis

Beyond aggregate-level interpretability, contemporary auditing standards increasingly mandate that automated classification systems provide instance-level transparency for each individual flagged transaction. To satisfy this requirement, the ExplainerAgent generates granular, localized feature importance attributions that collectively constitute a human-readable audit trail, enabling compliance officers and forensic auditors to examine the precise reasoning underlying each high-risk determination.

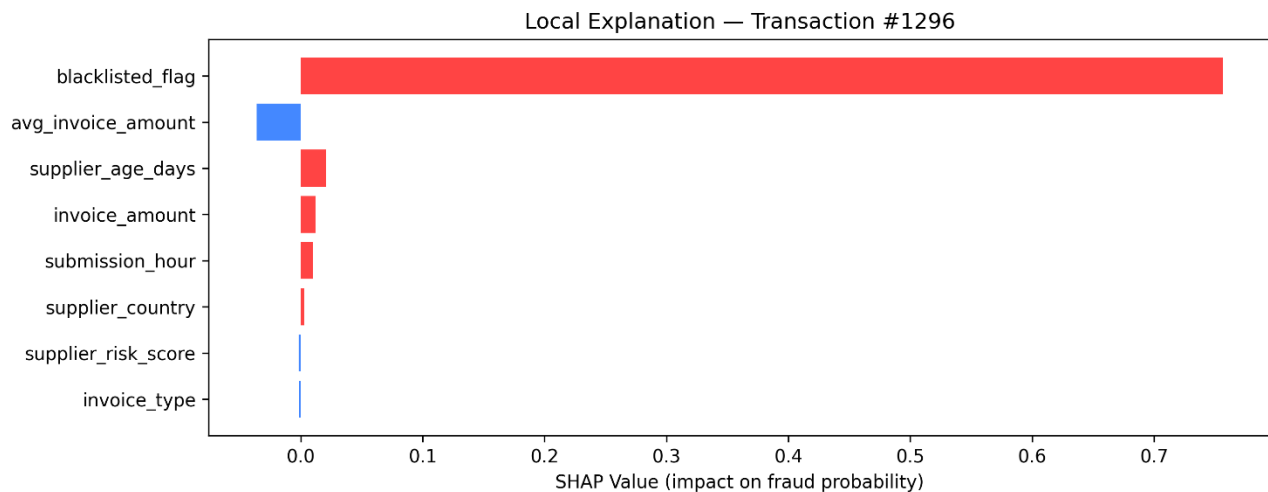


Fig. 4. Local SHAP Explanation for Transaction #1296 showing high-risk attribution.

A representative case is illustrated through Transaction #1296, as shown in Fig. 4, which received a hybrid risk score of 0.9894 placing it firmly within the highest echelon of the fraudulent classification boundary. The localized SHAP decomposition for this particular instance reveals that the `blacklisted_flag` variable served as the predominant risk driver, contributing an approximate positive shift of +0.75 toward the fraud probability estimate. This single feature attribution alone accounts for the majority of the elevated risk designation, indicating that the transaction originated from a supplier entity previously flagged under organizational compliance protocols. Analogous localized explanatory evidence was generated for additional high-risk cases, including Transaction #747 and Transaction #445, each presenting distinct yet equally interpretable feature contribution profiles. The systematic availability of such transaction-level justifications fulfills the foundational auditability requirement inherent to sensitive financial operations, wherein every automated decision must be defensible through explicit, traceable reasoning [18].

5.4. Feasibility and Decision Reconciliation

The collective body of experimental evidence presented throughout the preceding subsections substantiates the practical feasibility of deploying a hybrid multi-agent paradigm within realistic procurement fraud detection scenarios. The OrchestratorAgent demonstrated consistent and reliable execution of the decision fusion procedure through the Dynamic Weighted Fusion reconciliation mechanism, ensuring that deterministic rule violations identified by the RuleAgent were faithfully preserved within the composite hybrid score rather than being attenuated or overridden by the probabilistic outputs of the ensemble learner. This architectural property guarantees that transactions flagged on the basis of established regulatory criteria—such as supplier blacklist status or registration-to-invoice temporal anomalies—retain their elevated risk designation regardless of the statistical confidence assigned by the machine learning component. The resulting level of decision-making accountability is directly comparable to recent applications of explainable artificial intelligence developed for deployment within stringently regulated operational domains, including digital taxation administration [20]. Moreover, the integration of multi-layered explainability mechanisms—spanning both global feature impact analysis and localized transaction-level audit trails—ensures that the proposed system adequately satisfies the progressively demanding transparency standards prescribed within modern information engineering scholarship [23]. By furnishing explicit, human-comprehensible interpretive evidence alongside demonstrably strong predictive performance, the framework occupies a position at the intersection of computational intelligence and institutional accountability. This dual commitment to analytical rigor and interpretive openness aligns with the expanding interdisciplinary investigative trajectories envisioned under the contemporary paradigm of Explainable Artificial Intelligence 2.0, which advocates for the development of AI systems that are not merely accurate but fundamentally understandable, auditable, and contestable by human stakeholders [23].

Regarding operational scalability, the current distributed multi-agent architecture is inherently designed to support horizontal scaling. In large-scale enterprise deployments, the functionally distinct agents (DataAgent, MAgent, RuleAgent) can be deployed as independent, asynchronous microservices. This decoupling allows for real-time parallel processing of high-velocity transaction streams, preventing structural bottlenecks that typically hinder monolithic machine learning models.

6. Conclusion and Future Work

The present study has successfully demonstrated a rigorous proof of concept for a Hybrid Multi-Agent System expressly engineered to address the pervasive challenge of procurement fraud within organizational financial ecosystems. Through the deliberate architectural integration of probabilistic ensemble intelligence with deterministic regulatory enforcement logic, the proposed framework directly confronts the interpretive opacity that has long characterized conventional machine learning approaches to anomaly detection. The empirical findings obtained throughout the experimental phase substantiate the viability of this hybrid paradigm, with the system achieving a superior F1-Score of 0.8518 alongside a recall of 0.8190—collectively affirming its capacity to reliably identify complex fraudulent patterns embedded within inherently imbalanced financial transaction datasets. The principal scholarly contribution advanced by this research resides in the successful realization of a dual-layered, explicitly traceable audit mechanism that effectively reconciles the gap between algorithmic inference and human-interpretable transparency. Through the coordinated operation of the OrchestratorAgent and ExplainerAgent, the proposed system synthesizes statistical explanations grounded in SHAP-based feature attribution with verification outcomes derived from deterministic business rule evaluation. This architectural synergy is designed to support the generation of fraud alerts that are theoretically defensible from a regulatory standpoint, providing a traceable audit trail linked to identifiable transactional risk parameters. However, it must be emphasized that this alignment constitutes a theoretical and architectural capability. Formal certification and rigorous real-world compliance testing by accredited auditing bodies remain necessary before claiming definitive statutory compliance. The detailed examination of Transaction #1296 serves as an illustrative exemplification of this capability, wherein a markedly elevated hybrid risk score of 0.9894 was systematically deconstructed into a structured explanatory narrative encompassing both financial deviation indicators and explicit supplier blacklist violations. This degree of granular interpretive detail is of critical importance within the domain of forensic procurement auditing, as it ensures that no transaction undergoes automated flagging in the absence of a verifiable, regulation-aligned justification that can withstand institutional and external scrutiny.

Despite these promising outcomes, this study acknowledges several operational limitations. First, the dual-ensemble architecture introduces computational overhead that may require optimization for real-time processing in high-frequency transaction environments. Second, the deterministic rules maintained by the RuleAgent necessitate periodic manual review and updates by compliance officers to prevent rule decay against evolving fraud typologies. Furthermore, while the model mitigates internal bias through fixed business logic, its resilience against sophisticated adversarial attacks remains an area for future empirical validation.

Notwithstanding the promising outcomes established by this foundational investigation, several substantive avenues for methodological expansion and refinement warrant consideration in subsequent research efforts. A primary direction for future work involves scaling the proposed framework to accommodate high-velocity, real-time transaction streams characteristic of large-scale enterprise procurement environments, thereby transitioning the system from batch-oriented evaluation toward continuous monitoring capability. Additionally, broadening the knowledge repository maintained by the RuleAgent to incorporate regulatory parameters spanning diverse legal and tax jurisdictions would substantially enhance the geographical and institutional applicability of the detection framework. A further avenue of considerable promise concerns the prospective integration of Large Language Models into the explainability pipeline, which presents a significant opportunity to automate the transformation of raw feature attribution outputs into professionally structured audit reports suitable for direct consumption by compliance officers and regulatory bodies. This envisioned progression aligns with the broader intellectual trajectory articulated under the paradigm of Explainable Artificial Intelligence 2.0, which ultimately aspires to alleviate the cognitive burden imposed upon human auditors while simultaneously fortifying the procedural integrity and institutional accountability of organizational procurement governance mechanisms.

Author Contributions Statement

Lintang Arif Setianda – Conceptualization, Methodology, Data Curation, Software Implementation, Model Training, Formal Analysis, and Writing – Original Draft: Proposed research ideas, constructed the overall framework, handled dataset preprocessing, implemented the hybrid multi-agent model, performed in-depth analysis of experimental results using SHAP, and drafted the initial manuscript.

All authors have read and agreed to the published version of the manuscript.

Conflict of Interest Statement

The authors declare no conflicts of interest.

Funding Declaration

This research received no external funding.

Data Availability Statement

This study analyzed publicly available datasets. The Procurement Invoice Fraud Dataset v1 can be found here: <https://www.kaggle.com/datasets/tokelomashile2/procurement-invoice-fraud-dataset>. To ensure full reproducibility, the complete Python source code, experimental scripts, and parameters used in this study have been made publicly available for transparent community review and further research development in the author's dedicated GitHub repository at https://github.com/Lintangarif/Procurement-fraud-MAS/blob/main/hybrid_mas_procurement.ipynb.

Ethical Declarations

Not applicable. This study did not involve human subjects and/or animals.

Acknowledgments

The authors sincerely thank the Institut Teknologi Tangerang Selatan (ITTS) for providing the academic environment that supported this research. We also thank the reviewers for their professional evaluation and valuable recommendations, which have contributed to improving the quality of this manuscript.

Declaration of Generative AI in Scholarly Writing

The authors used generative AI technologies strictly to improve the language, readability, and formatting of the writing. After using these tools, the authors carefully reviewed and edited the generated results. The authors take full and sole responsibility for the final content of this publication.

Abbreviations

The following abbreviations are used in this manuscript:

AI - Artificial Intelligence
ML - Machine Learning
MAS - Multi-Agent System
XAI - Explainable Artificial Intelligence
SHAP - SHapley Additive exPlanations
ERP - Enterprise Resource Planning

References

- [1] ACFE, "Occupational Fraud 2024: A Report To The Nations®," *Assoc. Certif. Fraud Exam.*, pp. 1–106, 2024.
- [2] T. Pourhabibi, K. L. Ong, B. H. Kam, and Y. L. Boo, "Fraud detection: A systematic literature review of graph-based anomaly detection approaches," *Decis. Support Syst.*, vol. 133, p. 113303, 2020, doi: 10.1016/j.dss.2020.113303.
- [3] W. Hilal, S. A. Gadsden, and J. Yawney, "Financial Fraud: A Review of Anomaly Detection Techniques and Recent Advances," *Expert Syst. Appl.*, vol. 193, p. 116429, 2022, doi: 10.1016/j.eswa.2021.116429.
- [4] W. Saeed and C. Omlin, "Explainable AI (XAI): A systematic meta-survey of current challenges and future opportunities," *Knowledge-Based Syst.*, vol. 263, p. 110273, 2023, doi: 10.1016/j.knosys.2023.110273.
- [5] European Union, "Regulation (EU) 2024/1689 of the European Parliament and of the Council of 13 June 2024 laying down harmonised rules on artificial intelligence (Artificial Intelligence Act)," *Off. J. Eur. Union*, L168, vol. 1689, no. 3, pp. 1–153, 2024, [Online]. Available: <https://eur-lex.europa.eu/eli/reg/2024/1689/oj>
- [6] I. D. Raji *et al.*, "Closing the AI accountability gap: Defining an end-to-end framework for internal algorithmic auditing," *FAT* 2020 - Proc. 2020 Conf. Fairness, Accountability, Transpar.*, pp. 33–44, 2020, doi: 10.1145/3351095.3372873.
- [7] J. Nicholls, A. Kuppa, and N. A. Le-Khac, "Financial cybercrime: A comprehensive survey of deep learning approaches to tackle the evolving financial crime landscape," *IEEE Access*, vol. 9, pp. 163965–163986, 2021, doi: 10.1109/ACCESS.2021.3134076.

- [8] A. Barredo Arrieta *et al.*, “Explainable Artificial Intelligence (XAI): Concepts, taxonomies, opportunities and challenges toward responsible AI,” *Inf. Fusion*, vol. 58, pp. 82–115, 2020, doi: 10.1016/j.inffus.2019.12.012.
- [9] P. Linardatos, V. Papastefanopoulos, and S. Kotsiantis, “Explainable AI: A review of machine learning interpretability methods,” *Entropy*, vol. 23, no. 1, pp. 1–45, 2021, doi: 10.3390/e23010018.
- [10] W. Yang *et al.*, “Survey on Explainable AI: From Approaches, Limitations and Applications Aspects,” *Human-Centric Intell. Syst.*, vol. 3, no. 3, pp. 161–188, 2023, doi: 10.1007/s44230-023-00038-y.
- [11] V. Hassija *et al.*, “Interpreting Black-Box Models: A Review on Explainable Artificial Intelligence,” *Cognit. Comput.*, vol. 16, no. 1, pp. 45–74, 2024, doi: 10.1007/s12559-023-10179-8.
- [12] S. Ali *et al.*, “Explainable Artificial Intelligence (XAI): What we know and what is left to attain Trustworthy Artificial Intelligence,” *Inf. Fusion*, vol. 99, p. 101805, 2023, doi: 10.1016/j.inffus.2023.101805.
- [13] S. Asthana, “A Theoretical Framework for Hybrid Rule-Based and Machine-Learning Fraud Detection,” *Int. J. Comput. Trends Technol.*, pp. 7–13, Mar. 2026, doi: 10.14445/22312803/IJCTT-V74I3P102.
- [14] H. Verma and M. Sharma, “Rule-Based Hybrid AI – ML Models for Fraud Detection: A Systematic Review and Thematic Analysis,” *Int. J. Mod. Sci. Res. Technol.*, vol. 4, no. 1, 2026, doi: 10.5281/zenodo.18350970.
- [15] I. H. Sarker, “Machine Learning: Algorithms, Real-World Applications and Research Directions,” *SN Comput. Sci.*, vol. 2, no. 3, 2021, doi: 10.1007/s42979-021-00592-x.
- [16] T. Hu, “Financial Fraud Detection System Based on Improved Random Forest and Gradient Boosting Machine (GBM),” *Proc. - 2025 5th Asia-Pacific Conf. Commun. Technol. Comput. Sci. ACCTCS 2025*, pp. 761–765, 2025, doi: 10.1109/ACCTCS66275.2025.00137.
- [17] T. Mashile, “Procurement Invoice Fraud Dataset v1,” 2024. [Online]. Available: <https://www.kaggle.com/datasets/tokelomashile2/procurement-invoice-fraud-dataset>
- [18] D. Kaur, S. Uslu, K. J. Rittichier, and A. Durrresi, “Trustworthy Artificial Intelligence: A Review,” *ACM Comput. Surv.*, vol. 55, no. 2, 2023, doi: 10.1145/3491209.
- [19] X. Qu, “An Ensemble Learning Approach for Financial Fraud Detection: Combining Random Forest, K-Nearest Neighbors, and Gradient Boosting,” in *2025 5th International Conference on Artificial Intelligence, Big Data and Algorithms (CAIBDA)*, IEEE, Jun. 2025, pp. 1148–1154. doi: 10.1109/CAIBDA65784.2025.11182850.
- [20] Ł. Górski *et al.*, “Exploring explainable AI in the tax domain,” *Artif. Intell. Law*, vol. 33, no. 3, pp. 551–579, 2025, doi: 10.1007/s10506-024-09395-w.
- [21] M. Rakhshaninejad, M. Fathian, B. Amiri, and N. Yazdanjue, “An Ensemble-Based Credit Card Fraud Detection Algorithm Using an Efficient Voting Strategy,” *Comput. J.*, vol. 65, no. 8, pp. 1998–2015, Aug. 2022, doi: 10.1093/comjnl/bxab038.
- [22] V. Aworetan, “Next-generation Fraud Detection in U.S Financial Systems: Evaluating Hybrid AI and Rule-based Models for Real-time Threat Mitigation,” *World J. Adv. Res. Rev.*, vol. 23, no. 3, pp. 3317–3333, 2024, doi: 10.30574/wjarr.2024.23.3.3014.
- [23] L. Longo *et al.*, “Explainable Artificial Intelligence (XAI) 2.0: A manifesto of open challenges and interdisciplinary research directions,” *Inf. Fusion*, vol. 106, no. January, 2024, doi: 10.1016/j.inffus.2024.102301.
- [24] G. K. Awal and U. Tehlan, “New-Age Technologies to Combat Tax Evasion: Technological, Ethical, Legal, Social, and Economic (TELSE) Implications and a Scientometric Analysis,” *Int. J. Inf. Eng. Electron. Bus.*, vol. 17, no. 1, pp. 14–30, 2025, doi: 10.5815/ijieeb.2025.01.02.
- [25] V. Sri, S. Prakash, S. L. M. A. Ram, P. Nikhil, S. Dixt, and S. Sharma, “A Hybrid Multi-Model Approach for Credit Card Fraud Detection Using Diverse Datasets and Independent Feature Analysis,” *Int. J. Innov. Res. Technol.*, vol. 6, no. 12, pp. 1–5, 2024.

Authors' Profiles



Lintang Arif Setianda is currently pursuing the B.Sc. degree in Informatics at the South Tangerang Institute of Technology (ITTS), South Tangerang, Indonesia. He participated in the IOC Digital Hackathon AI/ML Data Analytics Upstream 2026 organized by SKK Migas and served as a featured speaker for the CROSS Webinar Series at ITTS, focusing on multi-agent systems. His major field of study is informatics, with research interests including multi-agent systems, neuro-symbolic AI, and Local Large Language Models (LLM).



Anas Nasrulloh, M.Kom., holds a Master's degree in Applied Computer Engineering from Universitas Budi Luhur. He is a lecturer in the Information Systems program at Institut Teknologi Tangerang Selatan. His research interests include data science, machine learning, and sentiment analysis using Python. This journal was written as a contribution to the advancement of artificial intelligence research and text data analysis in Indonesia.

Email: anas@itts.ac.id

How to cite this paper: Lintang Arif Setianda, Anas Nasrulloh, "A Multi-Agent System for Procurement Fraud Detection Integrating Rule-Based and Ensemble Learning with Explicit Audit Trail", International Journal of Information Engineering and Electronic Business(IJIEEB).